

Andrés Gaona

Mechatronics Engineer | Robotics R&D Engineer

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Professional Summary

Robotics R&D engineer with 8+ years of experience building real-time systems for mobile robots, legged robots, industrial automation, and vision-guided manipulation. Strong background in robot kinematics and dynamics, motion planning, position and torque control, ROS, EtherCAT, C++/Python, perception, and simulation-driven validation with NVIDIA Isaac tools. Experienced leading engineering teams, integrating software-control-hardware stacks, and moving robotic systems from prototype to field deployment.

Technical Skills

Robotics & Control	Robot kinematics and dynamics, motion planning, trajectory generation, real-time control, position and torque control, MPC, WBC, reinforcement learning, behavior trees, system integration
Robot Software	ROS, C++, Python, C, Linux, Git, Moveit, OpenCV, Pinocchio, Matlab
Motion & Hardware	EtherCAT, servo control, multi-axis synchronization, industrial communication, actuator integration, sensor feedback, embedded motion systems
Autonomy & Perception	SLAM, odometry, navigation, teleoperation, computer vision, object detection, object tracking, depth localization, grasp estimation
Robot Platforms	6-DOF robotic arms, quadruped robots, mobile robot platforms
Simulation & AI	NVIDIA Isaac Sim, Isaac Lab, sim-to-real validation, optimization algorithms

Work Experience

Project Leader / R&D Engineer

2024 – Present

Industrial Technology Research Institute (ITRI) | Taiwan

- Lead a team of 4 engineers developing autonomous mobile robot systems for hospital logistics and material dispatching.
- Architected a ROS2 and EtherCAT real-time control stack for reliable multi-actuator coordination across mobile robotic platforms.
- Designed semantic mapping pipelines using large language models to support high-level environment understanding and task execution.
- Developed quadruped control frameworks combining MPC for motion planning, WBC for coordinated actuation, and reinforcement learning for locomotion policy optimization.
- Validated robotic systems in NVIDIA Isaac Sim and Isaac Lab, reducing risk before sim-to-real deployment.
- Led integration across perception, control, and hardware teams to deliver end-to-end robotic system functionality.

Development & Research Engineer

2018 – 2024

Industrial Technology Research Institute (ITRI) | Taiwan

- Developed and maintained MCCL, a high-performance EtherCAT-based C++ control library for industrial robotic systems.
- Implemented real-time communication and control loops for multi-motor platforms, including Dynamixel-based systems.
- Delivered custom robotics and automation solutions for recycling systems, metal forming, and industrial assembly lines.
- Owned software lifecycle practices including version control, debugging, deployment support, and system-level optimization.
- Provided on-site troubleshooting for industrial deployments and mentored junior engineers in robotic control, EtherCAT communication, and system integration.

Research Associate

2015 – 2016

University of the Armed Forces ESPE, Mechanical & Mechatronics Engineering Department | Ecuador

- Contributed to robotics research and experimental system development, including methodology design, validation, and technical documentation.
- Supported peer-reviewed publications through data analysis, experimental setup improvement, and technical writing.
- Provided academic tutoring in control systems and mechatronics.

Selected Projects

- **Quadruped Robot Locomotion & Control Framework**
 - Developed a torque-controlled four-legged robot stack combining MPC, WBC, reinforcement learning, state machines, and smooth motion interpolation.
 - Validated locomotion and adaptive behaviors in NVIDIA Isaac Sim and Isaac Lab, including walking, trotting, balance recovery, and climbing.
- **Real-Time Multi-Vendor EtherCAT Motion Control Platform**
 - Developed MCCL, a C++ real-time motion control platform for deterministic multi-axis synchronization across heterogeneous EtherCAT devices.
 - Built vendor abstraction for CoE objects, CSP/PP motion modes, servo drives, and ESI-to-ENI device on-boarding.
- **Autonomous Mobile Robot (AMR) for Indoor & Hospital Logistics**
 - Built a ROS-based AMR platform for indoor hospital material dispatching, integrating navigation, odometry, teleoperation, semantic mapping, and control.
 - Enabled real-time EtherCAT communication for reliable actuator control and mobile platform coordination.
- **Deep Learning Robotic Plastic Waste Sorter**
 - Developed an industrial robotic sorter combining YOLOv4, GPU inference, object tracking, depth localization, and delta robot control.
 - Synchronized vision and motion pipelines with MODBUS/EtherCAT for real-time detection, classification, picking, and sorting.
- **Vision-Guided Robotic Manipulation & Path Generation**
 - Planned trajectory generation for 6-DOF robotic arms, including spray-painting and point-cloud-based automatic path generation.
 - Implemented vision-guided pick-and-place with SCARA robots, EtherCAT servo control, object detection, orientation estimation, and grasping.
- **Smart Tuning & Optimization for Motion Control**
 - Applied PSO and sensor feedback for controller and servo-driver tuning across robotic and industrial motion platforms.
 - Built digital twin, AR/VR robot teaching, synchronous gantry control, Stewart platform, and predictive maintenance systems.

Education

Master in Mechanical and Automation Systems 2016 – 2018

National Taipei University of Technology (NTUT) | Taiwan

- **Thesis:** Integration of Convolutional Neural Networks (CNNs) and Particle Swarm Optimization (PSO) for robotic grasp estimation.
- **Awards:** Master Thesis Excellence Award (2019), Best Student Award (2016), Excellence Scholarship (2016–2017).

Mechatronics Engineering 2009 – 2015

University of The Armed Forces ESPE | Sangolqui, Ecuador

- **Thesis:** Mobile robotics for Simultaneous Localization and Mapping (SLAM).
- **Award:** High-Performance Athlete Scholarship (2009–2015).

Languages

Spanish	Native
English	Advanced (C1)
Mandarin	Intermediate (B1)